Laravel Authentication Lab

Completed Solution • Laravel 12 • IT Student Portal

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Project overview

This solution completes the lab requirements: registration with validation, hashed password storage, login with Laravel session authentication, a protected dashboard, logout, and a test checklist. Create the files at the exact paths shown below inside your Laravel 12 project.

1. Project setup

Create a project only if you do not already have one:

```
composer create-project laravel/laravel it-auth-lab "^12.0"
cd it-auth-lab
```

Create an empty MySQL database named **it_auth_lab**. Configure `.env` (use your own database username/password if needed):

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=it_auth_lab
DB_USERNAME=root
DB_PASSWORD=
SESSION_DRIVER=file
```

```
php artisan config:clear
php artisan migrate
php artisan make:controller AuthController
php artisan serve
```

2. Named routes — routes/web.php

```
<?php

use App\\Http\\Controllers\\AuthController;
use Illuminate\\Support\\Facades\\Route;

Route::redirect('/', '/login');

Route::middleware('guest')->group(function () {
    Route::view('/register', 'auth.register')->name('register');
    Route::post('/register', [AuthController::class, 'register'])
        ->name('register.store');

    Route::view('/login', 'auth.login')->name('login');
    Route::post('/login', [AuthController::class, 'login'])
        ->name('login.store');
});

Route::middleware('auth')->group(function () {
    Route::get('/dashboard', function () {
        return view('dashboard');
    })->name('dashboard');

    Route::post('/logout', [AuthController::class, 'logout'])
        ->name('logout');
});
```

3. Registration form — resources/views/auth/register.blade.php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Create an account</title>
</head>
<body>
    <h1>Create an account</h1>

    @if ($errors->any())
        <ul>
            @foreach ($errors->all() as $error)
                <li>{{ $error }}</li>
            @endforeach
        </ul>
    @endif

    @if (session('success'))
        <p>{{ session('success') }}</p>
    @endif

    <form method="POST" action="{{ route('register.store') }}">
        @csrf

        <label for="name">Full name</label>
        <input id="name" name="name" type="text"
               value="{{ old('name') }}" required>

        <label for="email">Email</label>
        <input id="email" name="email" type="email"
               value="{{ old('email') }}" required>

        <label for="password">Password (at least 8 characters)</label>
        <input id="password" name="password" type="password"
               minlength="8" required>

        <label for="password_confirmation">Confirm password</label>
        <input id="password_confirmation" name="password_confirmation"
               type="password" minlength="8" required>

        <button type="submit">Register</button>
    </form>

    <a href="{{ route('login') }}">Already have an account?</a>
</body>
</html>
```

4. Login form — resources/views/auth/login.blade.php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Log in</title>
</head>
<body>
    <h1>Log in</h1>

    @if ($errors->any())
        <ul>
            @foreach ($errors->all() as $error)
                <li>{{ $error }}</li>
            @endforeach
        </ul>
    @endif

    @if (session('success'))
        <p>{{ session('success') }}</p>
    @endif

    <form method="POST" action="{{ route('login.store') }}">
        @csrf

        <label for="email">Email</label>
        <input id="email" name="email" type="email"
               value="{{ old('email') }}" required>

        <label for="password">Password</label>
        <input id="password" name="password" type="password" required>

        <button type="submit">Login</button>
    </form>

    <a href="{{ route('register') }}">Create an account</a>
</body>
</html>
```

5. Controller — app/Http/Controllers/AuthController.php

```
<?php

namespace App\\Http\\Controllers;

use App\\Models\\User;
use Illuminate\\Http\\Request;
use Illuminate\\Support\\Facades\\Auth;
use Illuminate\\Support\\Facades\\Hash;

class AuthController extends Controller
{
    public function register(Request $request)
    {
        $data = $request->validate([
            'name' => 'required|string|max:255',
            'email' => 'required|email|max:255|unique:users,email',
            'password' => 'required|string|min:8|confirmed',
        ]);

        $user = new User();
        $user->name = $data['name'];
        $user->email = $data['email'];
        $user->password = Hash::make($data['password']);
        $user->save();

        return redirect()->route('login')
            ->with('success', 'Account created. Please log in.');
    }

    public function login(Request $request)
    {
        $credentials = $request->validate([
            'email' => 'required|email',
            'password' => 'required|string',
        ]);

        $accepted = Auth::attempt($credentials);
```

```
        if (! $accepted) {
            return back()->withErrors([
                'email' => 'Email or password is incorrect.',
            ])->onlyInput('email');
        }

        $request->session()->regenerate();

        return redirect()->intended(route('dashboard'));
    }

    public function logout(Request $request)
    {
        Auth::logout();
        $request->session()->invalidate();
        $request->session()->regenerateToken();

        return redirect()->route('login');
    }
}
```

6. Protected dashboard — resources/views/dashboard.blade.php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>IT Student Portal</title>
</head>
<body>
    <h1>IT Student Portal</h1>
    <p>Welcome, {{ auth()->user()->name }}.</p>
    <p>Email: {{ auth()->user()->email }}</p>

    <form method="POST" action="{{ route('logout') }}">
        @csrf
        <button type="submit">Logout</button>
    </form>
</body>
</html>
```

7. Verify the routes and test the portal

```
php artisan route:list
php artisan route:clear
```

Test	Expected result	Result
/dashboard while logged out	Redirects to login	Pass / Fix
Register valid, new details	One user created; success message	Pass / Fix
Register duplicate email	Validation error; no extra user	Pass / Fix
Password shorter than 8 characters	Validation error; no account	Pass / Fix
Password confirmation mismatch	Validation error; no account	Pass / Fix
Correct login	Dashboard shows current user's name/email	Pass / Fix
Incorrect password	Error; dashboard stays protected	Pass / Fix
/login while already logged in	Guest middleware redirects to dashboard	Pass / Fix
Logout then open /dashboard	Redirects to login	Pass / Fix
Login with second test account	Second account's details appear	Pass / Fix

8. Short answers for the instructor

1. Why use Hash::make()?
It hashes the password before storing it. The database should not contain the plain-text password.

2. How does auth middleware protect the dashboard?
It checks whether the visitor is authenticated. If the visitor is not signed in, Laravel redirects them to the login page.

3. What does @csrf add to a form?
It includes a CSRF token so Laravel can verify that a POST request came from a valid form session and help protect against cross-site request forgery.

Important notes

• Keep passwords empty when forms are redisplayed; do not use old('password').
• Do not store password_confirmation in the users table.
• Every POST form, including logout, must include @csrf.
• The dashboard shows the account currently signed in; the name entered during registration is displayed dynamically.